CLINICAL · MOUTH & THROAT

Tonsillitis and peritonsillar abscess

The tonsils are the most operated-on lymphoid tissue in the body and among the most misunderstood. Most sore throats need no antibiotic and most tonsils need no surgery; but the patient who cannot swallow his own saliva, whose uvula is pushed across and whose jaw will not open has a quinsy that must be drained, and a child with a membrane on the tonsils and a bull neck may have diphtheria. Know the tonsil, the infections and the operation, and the dangerous throat stands out from the ordinary one.

AT A GLANCE

- Sore throat with trismus means quinsy.
- Drain and treat, watching the airway.
- Use Paradise criteria for tonsillectomy.
- An asymmetric adult tonsil needs biopsy.
- Intracapsular tonsillectomy means less pain and bleeding; codeine is contraindicated under 18 for OSA.

1 CHAPTER ONE Applied anatomy

The palatine tonsil.

1.1 Position and surfaces

Each tonsil is an ovoid mass of lymphoid tissue in the lateral wall of the oropharynx, in the *tonsillar fossa* between the anterior (palatoglossal) and posterior (palatopharyngeal) pillars. It is larger than it looks, extending up into the soft palate, down into the base of tongue and forwards under the anterior pillar.

- **Medial surface:** non-keratinising stratified squamous epithelium dipping into 12 to 15 *crypts*; the largest, near the upper pole, is the *crypta magna* (intratonsillar cleft), the remnant of the second pharyngeal pouch. Crypts fill with cheesy debris that can be expressed by pressing on the anterior pillar.
- **Lateral surface:** a well-defined fibrous *capsule*, separated from the bed by loose areolar tissue: the plane of dissection at tonsillectomy and the space where pus

collects in a quinsy.

- **Upper pole:** reaches into the soft palate under a semilunar fold enclosing the *supratonsillar fossa*.
- **Lower pole:** attached to the tongue, separated from it by the *tonsillolingual sulcus*, a hidden site for carcinoma.

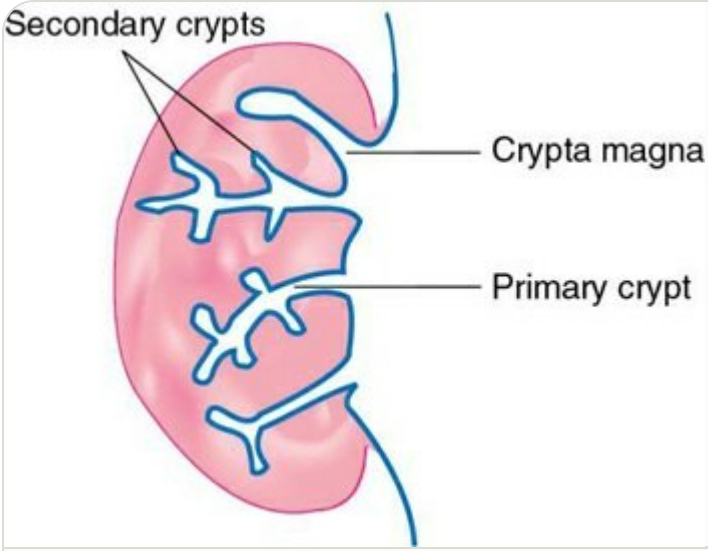


FIGURE 1 Primary and secondary crypts of the tonsil, with the crypta magna.
Dhingra, fig. 51.1

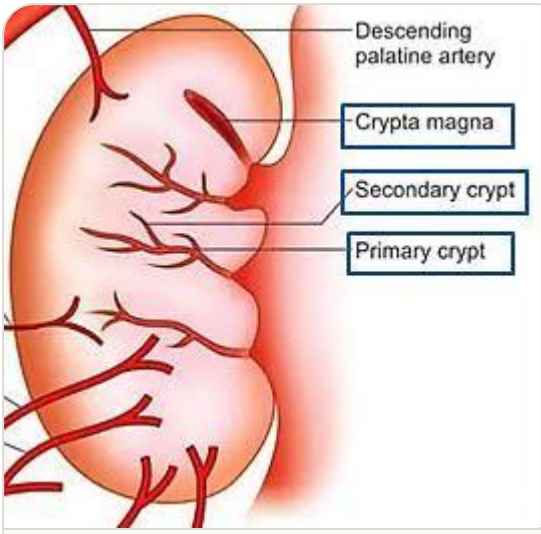


FIGURE 2 Tonsillar crypts and blood vessels.
Anatomy atlas · ENT Steps p. 214

1.2 Bed and relations

The bed is formed by the superior constrictor and styloglossus. The glossopharyngeal nerve and an elongated styloid process may lie in the lower part

of the fossa and can be reached through it after tonsillectomy. Beyond the constrictor lie the facial artery, submandibular gland, posterior belly of digastric, medial pterygoid and the angle of the mandible; the internal carotid artery lies about 2.5 cm behind and lateral to the tonsil.

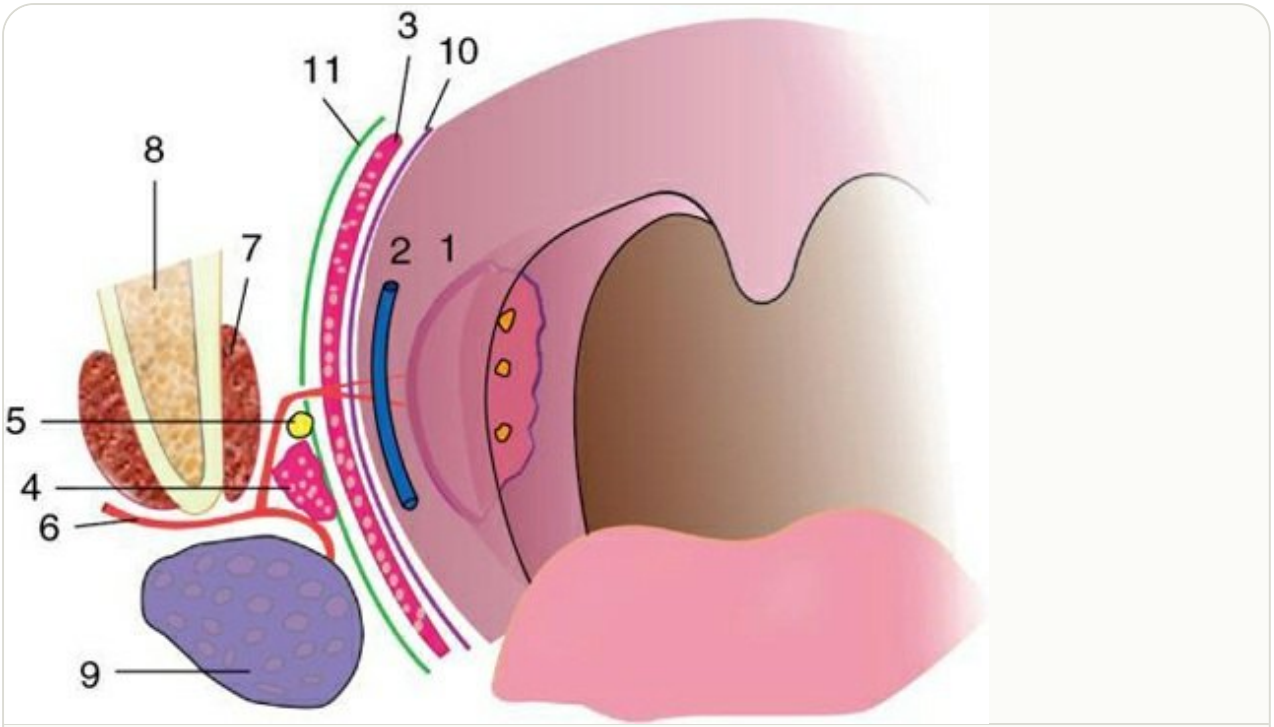


FIGURE 3 Relations of the tonsil: capsule (1), loose areolar tissue with the paratonsillar vein (2), superior constrictor (3), styloglossus (4), glossopharyngeal nerve (5), facial artery (6), medial pterygoid (7), angle of mandible (8), submandibular gland (9).

Dhingra, fig. 51.2

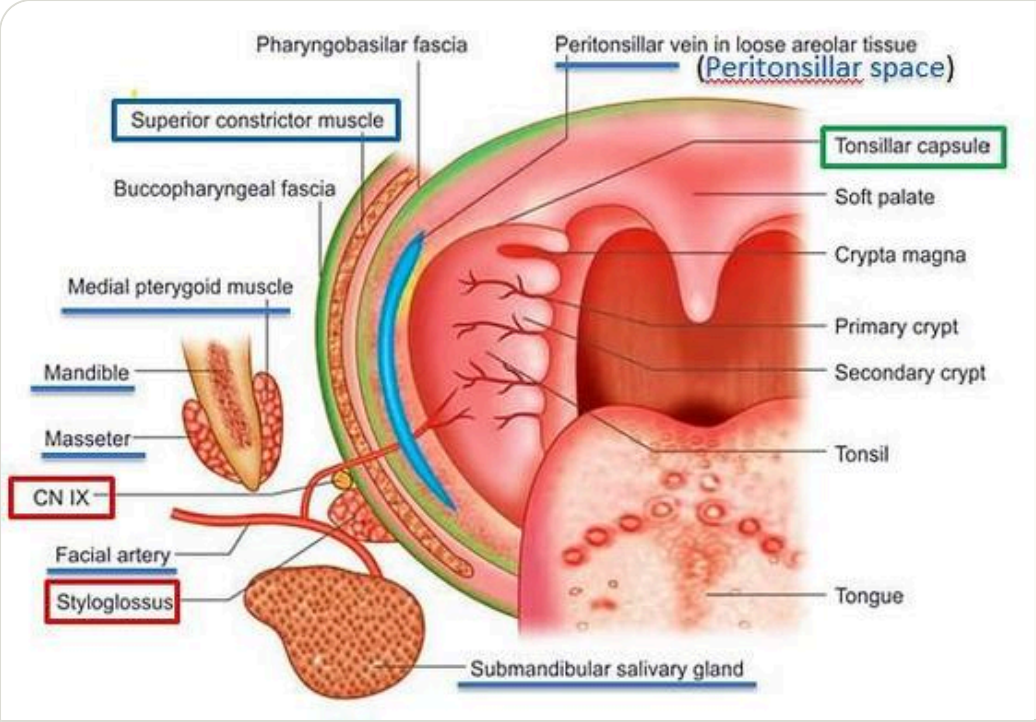


FIGURE 4 The bed of the tonsil in horizontal section.
Anatomy atlas · ENT Steps p. 215

1.3 Blood supply

Five arteries: the *tonsillar branch of the facial artery* (the main supply, entering at the lower pole), the ascending palatine (facial), the ascending pharyngeal, the dorsal lingual branches (lingual) and the descending palatine (maxillary). Veins drain to the *paratonsillar (external palatine) vein*, which descends from the soft palate across the lateral surface of the capsule to the common facial vein and pharyngeal plexus; it is the usual source of bleeding at tonsillectomy.

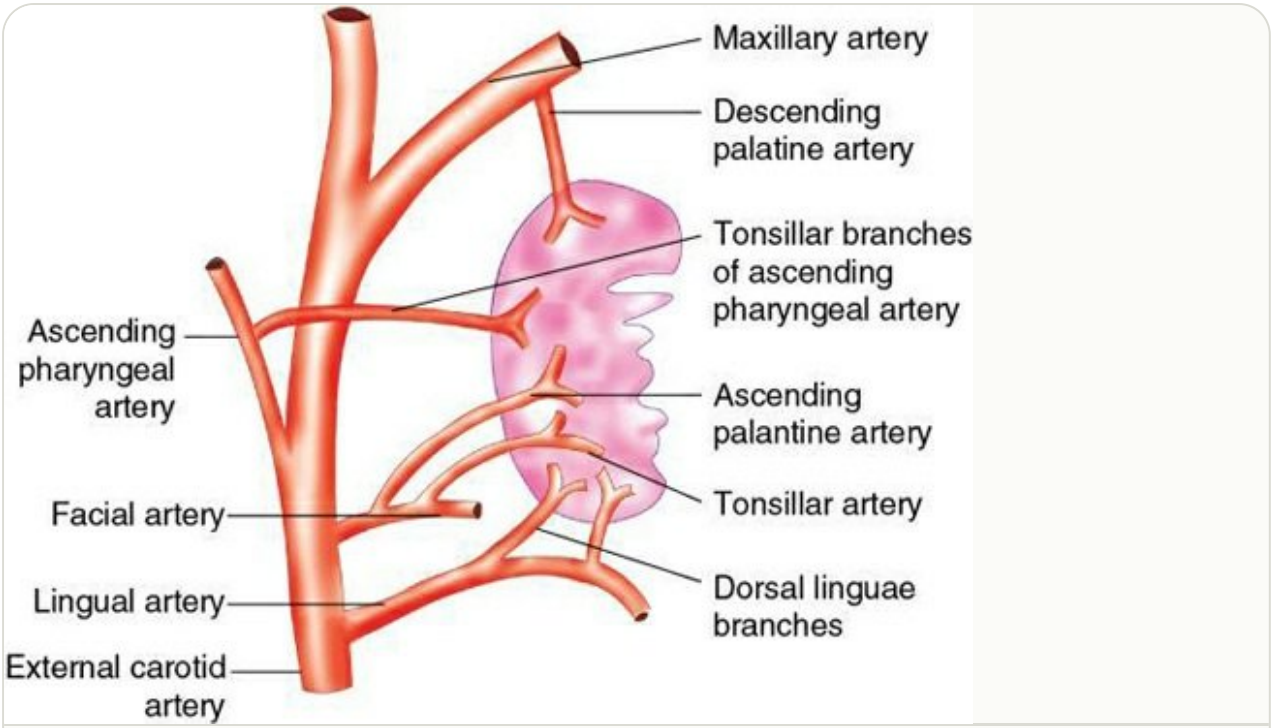


FIGURE 5 Arterial supply of the tonsil.
Dhingra, fig. 51.3

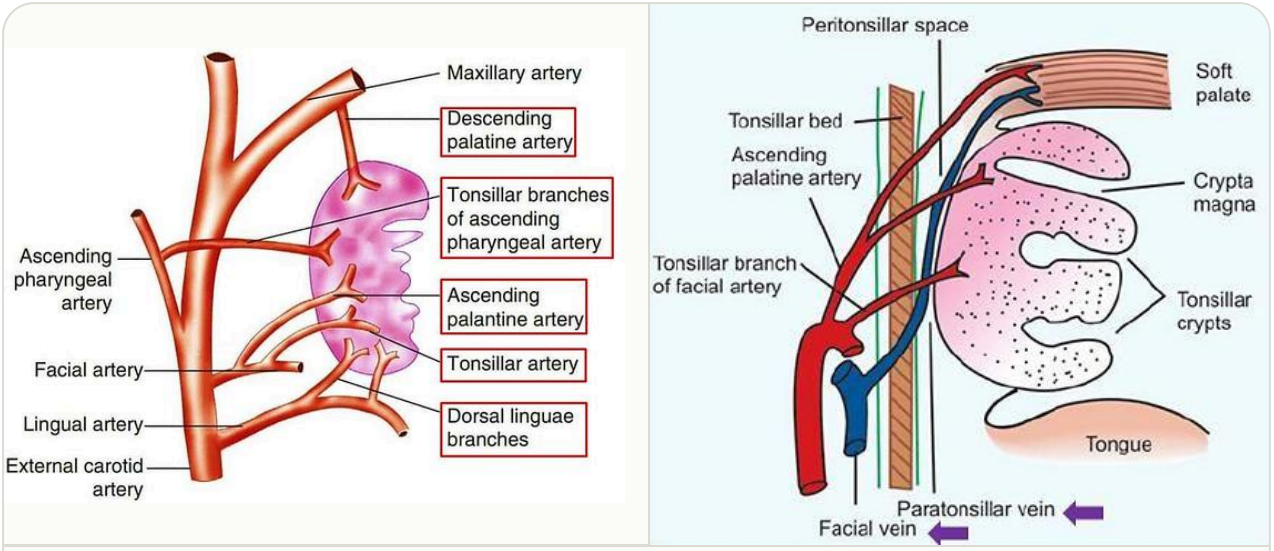


FIGURE 6 Arterial supply of the palatine tonsil, and the paratonsillar vein with the tonsillar arteries.
Anatomy atlas · ENT Steps p. 217, ENT Steps p. 218

1.4 Lymph and nerves

Lymphatics pierce the superior constrictor to the *jugulodigastric (tonsillar) node* below the angle of the mandible. Sensation comes from the glossopharyngeal nerve

and lesser palatine nerves; hence the earache of tonsillitis and after tonsillectomy.

1.5 Function

The crypts and their specialised epithelium (M cells, antigen-presenting cells) bring inhaled and swallowed antigens to B-cell follicles that make antibody, and alert the wider immune system. The tonsils are most active between four and ten years and involute after puberty. Removal has not been shown to impair immunity, but should still be done only for a clear indication.

2 CHAPTER TWO Acute tonsillitis

The common infection.

2.1 Types

- **Catarrhal (superficial):** part of a generalised, usually viral, pharyngitis.
- **Follicular:** pus in the crypts appears as yellow spots at their openings.
- **Parenchymatous:** the whole tonsil is uniformly swollen and red.
- **Membranous:** exudate from the crypts coalesces into a membrane over the surface.

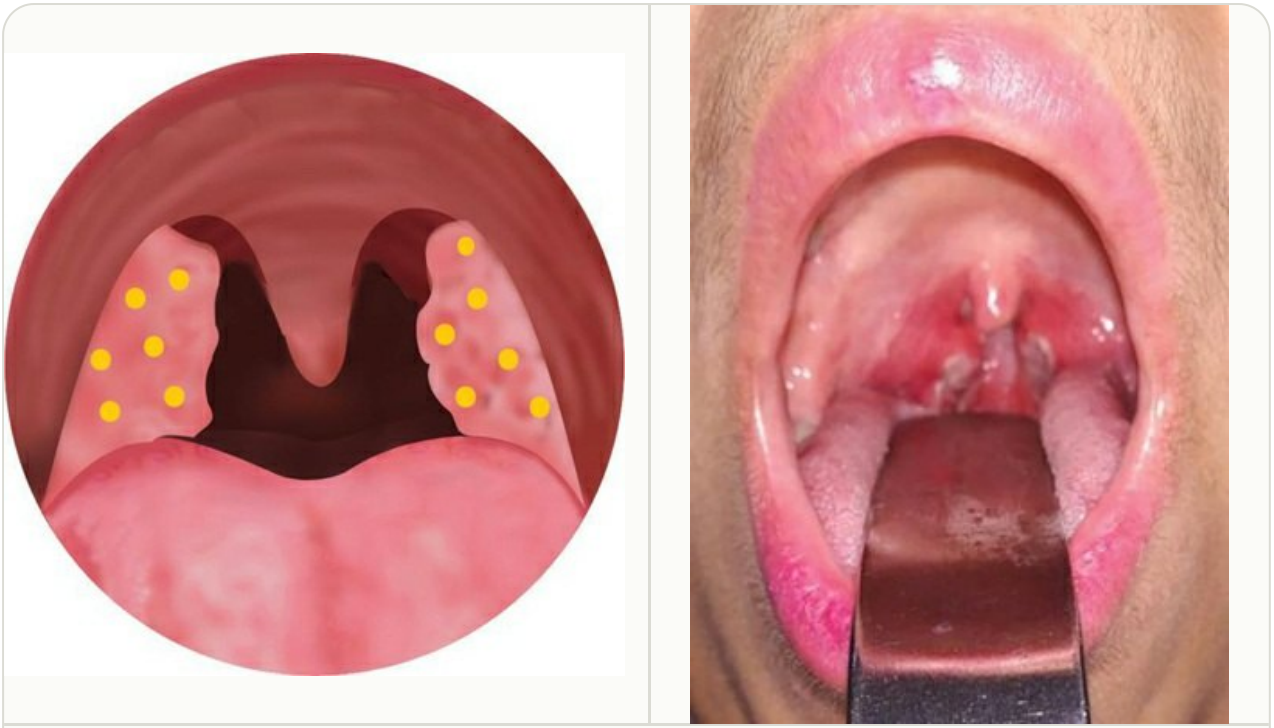


FIGURE 7 First, acute follicular tonsillitis. Second, acute membranous tonsillitis.
Dhingra, figs 51.4 and 51.5

2.2 Cause

School children mostly, and young adults; uncommon in infants and over fifty. Group A beta-haemolytic streptococcus is the commonest bacterial cause; staphylococci, pneumococci and *H. influenzae* also, often after a virus. Most episodes overall are viral.

2.3 Clinical features

Sore throat, painful swallowing (a child may refuse food), fever of 38 to 40°C with rigors, referred earache or true otitis media, malaise, and in children abdominal pain from mesenteric adenitis that can mimic appendicitis. The breath is foul, the tongue coated, the pillars and palate red; the tonsils are swollen with spots of pus or a membrane that wipes off easily; the jugulodigastric nodes are enlarged and tender.



FIGURE 8 Acute follicular tonsillitis: pus beads on the left tonsil; on the right they have coalesced into a membrane.

Dhingra, fig. 51.6

2.4 Treatment

Rest, fluids and analgesia (paracetamol or ibuprofen; aspirin is avoided in children). For bacterial tonsillitis, penicillin for ten days (erythromycin in allergy); amoxicillin is best avoided when glandular fever is possible.

2.5 Complications

- Chronic tonsillitis with recurrent attacks.
- Peritonsillar abscess, parapharyngeal abscess, cervical abscess from suppurating nodes.
- Acute otitis media.
- Rheumatic fever, acute glomerulonephritis, and endocarditis in valvular heart disease.

3 CHAPTER THREE
The membrane on the tonsil

A differential that matters.

MEMBRANE OVER THE TONSIL

CONDITION	CLUES
Membranous tonsillitis	Abrupt onset, high fever; membrane on the tonsil only, wipes off easily
Diphtheria	Slower onset, low fever, toxic child; dirty grey membrane spreading beyond the tonsil to palate and pharynx, adherent, bleeds when removed; bull neck
Vincent angina	Insidious, little fever; membrane usually on one tonsil, lifting off to leave an irregular ulcer; fusiform bacilli and spirochaetes on smear
Infectious mononucleosis	Young adult; both tonsils large and coated; posterior cervical nodes, spleen; atypical lymphocytes, positive Paul–Bunnell (Monospot) test
Agranulocytosis	Very ill; necrotic ulcers across the oropharynx; neutrophils profoundly reduced
Leukaemia	Anaemia, bleeding gums; blast cells in blood and marrow
Others	Aphthous ulcer, traumatic ulcer (membrane within 24 hours), candida, malignancy

Investigations: total and differential counts, blood film, throat swab and culture, Monospot test, marrow examination and biopsy as indicated.

3.1 Faucial diphtheria

Corynebacterium diphtheriae, spread by droplets, incubation two to six days; carriers exist. A grey-white tenacious membrane forms on the tonsils and spreads to the palate and posterior wall, bleeding when lifted; tender jugulodigastric nodes give a *bull neck*; the child is toxic though the fever is modest. The exotoxin damages the heart (myocarditis, arrhythmia, circulatory failure) and nerves (palatal, ocular and diaphragmatic paralysis, weeks later); laryngeal membrane obstructs the airway.

TREAT DIPHTHERIA ON SUSPICION

- Give **antitoxin** without waiting for culture: 20,000 to 40,000 units if under 48 hours and confined to the tonsils, 80,000 to 120,000 units if longer or more extensive, intravenously over an hour after a test dose, with adrenaline ready.
- **Penicillin or erythromycin** to kill the organism.
- **Isolate;** discharge after three negative swabs 24 hours apart; give a booster, since infection does not confer immunity.
- Immunise contacts and give them erythromycin for a week.

CHAPTER FOUR

4 Chronic tonsillitis

Infection that does not go away.

4.1 Causes

Incomplete resolution of acute attacks with walled-off microabscesses in the follicles, subclinical infection, and infected sinuses or teeth; mostly children and young adults.

4.2 Types

- **Chronic follicular:** crypts full of infected cheesy material, seen as yellow spots.
- **Chronic parenchymatous:** hyperplasia makes the tonsils very large; they may meet in the midline and obstruct speech, swallowing and breathing, with sleep apnoea and eventually cor pulmonale.
- **Chronic fibroid:** small, infected tonsils with recurrent sore throats.



FIGURE 9 Parenchymatous tonsillitis: the two tonsils almost touch.
Dhingra, fig. 51.7

4.3 Features

Recurrent sore throats, chronic irritation and cough, bad taste and halitosis, and with large tonsils thick speech, difficulty swallowing and choking at night. *Flushing of the anterior pillars* compared with the surrounding mucosa and persistent *enlargement of the jugulodigastric nodes* are reliable signs; pressure on the pillar may express pus.

TONSIL SIZE GRADING

Grade 0 tonsils are within the fossa; grade 1 occupy under 25% of the oropharyngeal airway; grade 2, 25 to 50%; grade 3, 50 to 75%; grade 4, more than 75% ("kissing" tonsils).

4.4 Local complications

- **Intratonsillar abscess:** a blocked crypt fills with pus; a red, swollen, very painful tonsil; antibiotics, drainage, and later tonsillectomy.
- **Tonsillolith:** a calcified crypt plug felt as a gritty mass, removed or treated by tonsillectomy.

- **Tonsillar cyst:** a yellow swelling from a blocked crypt, often symptomless, easily drained.



FIGURE 10 A tonsillar cyst on the left (arrow).
Dhingra, fig. 51.8

4.5 Lingual tonsil

Acute lingual tonsillitis gives a one-sided lump feeling and dysphagia; *compensatory hypertrophy* follows tonsillectomy and causes a lump sensation, thick voice and cough; a *lingual tonsil abscess* causes severe dysphagia, salivation, painful protrusion and risks laryngeal oedema.

4.6 Periodic fever (PFAPA)

In young children, recurrent episodes every three to eight weeks of fever, aphthous ulcers, pharyngitis and cervical adenitis lasting a few days, with complete health in between. Cyclic neutropenia must be excluded. A single dose of steroid aborts attacks; tonsillectomy is curative in most.

5 CHAPTER FIVE Peritonsillar abscess

Quinsy.

5.1 Definition and cause

Pus in the peritonsillar space between the tonsil capsule and the superior constrictor. Usually a crypt, often the crypta magna, becomes blocked and forms an intratonsillar abscess that bursts through the capsule; some arise from the mucous (Weber) glands of the supratonsillar fossa. Cultures are usually mixed aerobic (*Streptococcus pyogenes*, *Staphylococcus aureus*) and anaerobic.

5.2 Clinical features

Mostly young adults, usually one-sided.

- **General:** fever to 40°C, rigors, malaise, headache.
- **Local:** severe one-sided throat pain; odynophagia so marked that saliva dribbles from the mouth and the patient becomes dehydrated; a muffled *hot-potato voice*; foul breath; ipsilateral earache; and *trismus* from spasm of the medial pterygoid.

EXAMINATION FINDINGS IN QUINSY

SIGN	EXPLANATION
Bulging of the soft palate and anterior pillar above the tonsil	Pus in the supratonsillar part of the space
Uvula swollen and pushed to the opposite side	Mass effect
Tonsil buried in oedematous pillars, pushed down and medially	Displacement
Mucopus over the tonsil, tender jugulodigastric nodes	Infection
Neck tilted to the affected side (torticollis)	Muscle spasm



FIGURE 11 A left peritonsillar abscess.
Dhingra, fig. 52.4

TRISMUS IS THE KEY

Lee calls trismus the best single indicator of an abscess rather than peritonsillar cellulitis. If you are unsure, aspirate with a needle: pus settles the question and gives a culture.

5.3 Investigation

Diagnosis is clinical. Needle aspiration confirms pus. Contrast-enhanced CT when the diagnosis is uncertain, trismus prevents examination, or spread to the parapharyngeal space is suspected.

5.4 Treatment

1. Admit; intravenous fluids.
2. High-dose intravenous antibiotics covering aerobes and anaerobes (penicillin with metronidazole, or co-amoxiclav); a single dose of steroid reduces pain and trismus.
3. Analgesia; avoid aspirin. Saline or peroxide mouthwashes.
4. **Drain the pus:** needle aspiration, or incision at the point of maximum bulge, which lies just lateral to where a horizontal line through the base of the uvula meets a vertical line along the anterior pillar. A guarded blade makes a small stab and sinus forceps open the cavity; the forceps may be reinserted the next day.

5. **Tonsillectomy:** an *interval* tonsillectomy four to six weeks later is advised after a second quinsy in adults (Lee quotes recurrence of about 17%), and more readily in children or those with recurrent tonsillitis. *Quinsy (hot) tonsillectomy* during the acute illness is preferred by some, at the cost of more bleeding and the risk of rupture at induction.

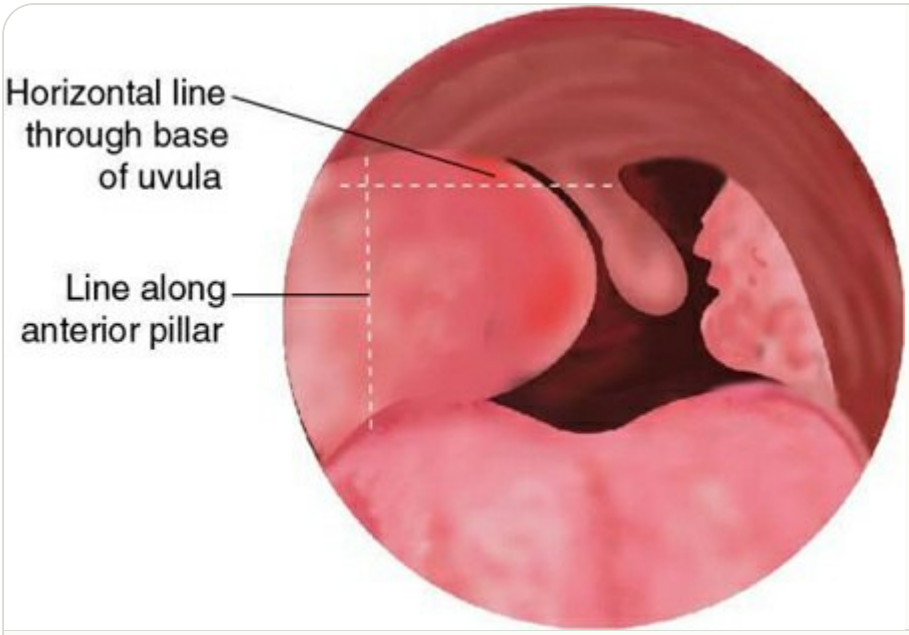
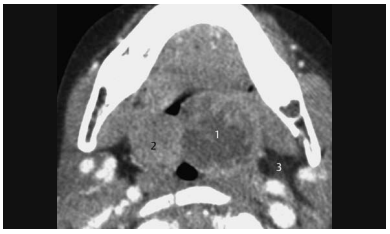


FIGURE 12 The site of drainage: just lateral to the junction of a vertical line along the anterior pillar and a horizontal line through the base of the uvula.

Dhingra, fig. 52.5

IMAGING IN THE APP 1 case

A tonsillar abscess on imaging.



Tonsillar Abscess

Infection & inflammation · 7 images

5.5 Complications

Parapharyngeal abscess (a quinsy is a potential parapharyngeal abscess), laryngeal oedema needing tracheostomy, aspiration pneumonia or lung abscess after spontaneous rupture, septicaemia with endocarditis, nephritis or brain abscess, internal jugular vein thrombosis (Lemierre syndrome), and haemorrhage from erosion of the carotid or jugular vessels.

6 CHAPTER SIX Tonsillectomy

Indications, technique and complications.

6.1 Indications

6.1.1 Absolute (Dhingra)

- **Recurrent throat infection:** seven or more episodes in one year, five a year for two years, or three a year for three years (the Paradise criteria), or two weeks or more of school or work lost in a year.
- **Peritonsillar abscess:** in children four to six weeks after treatment; in adults after a second attack.
- **Tonsillitis causing febrile seizures.**
- **Hypertrophy** causing obstructive sleep apnoea, difficulty swallowing or interference with speech.
- **Suspicion of malignancy:** a one-sided enlarged tonsil may be lymphoma in a child or squamous carcinoma in an adult; excisional biopsy. Also in the search for an unknown primary.

6.1.2 Relative

Diphtheria or streptococcal carriers unresponsive to antibiotics, chronic tonsillitis with halitosis unresponsive to treatment, and recurrent streptococcal tonsillitis in valvular heart disease.

6.1.3 As part of another operation

Uvulopalatopharyngoplasty, glossopharyngeal neurectomy, and removal of an elongated styloid process.

CURRENT PRACTICE

Guidelines now recommend watchful waiting for recurrent sore throat that falls short of the Paradise criteria, and give obstructive sleep-disordered breathing as the commonest indication in children. A single intraoperative dose of dexamethasone reduces vomiting; routine postoperative antibiotics are not recommended; codeine is avoided in children.

6.2 Contraindications

Haemoglobin below 10 g/dL; acute upper respiratory infection, including acute tonsillitis; children under three (relative); overt or submucous cleft palate; bleeding disorders (von Willebrand disease, haemophilia, leukaemia, purpura, aplastic anaemia); uncontrolled diabetes, hypertension, heart disease or asthma; and, traditionally, during an epidemic of polio and during menstruation.

6.3 Technique (dissection and snare)

1. General anaesthesia with a tube; *Rose's position* (head extended on a shoulder roll, head ring), avoiding hyperextension.
2. The Boyle–Davis gag is opened and held by Draffin's bipods.
3. The tonsil is grasped and drawn medially; the mucosa is incised where it reflects from the tonsil onto the anterior pillar, and the incision carried round the upper pole.
4. The upper pole is freed with blunt scissors or a dissector in the plane outside the capsule, and dissection continues downwards to the lower pole.
5. A snare is passed over the pedicle at the lower pole, tightened and the tonsil removed.
6. A swab is pressed into the fossa; bleeding points are tied or coagulated. The other side is done in the same way.

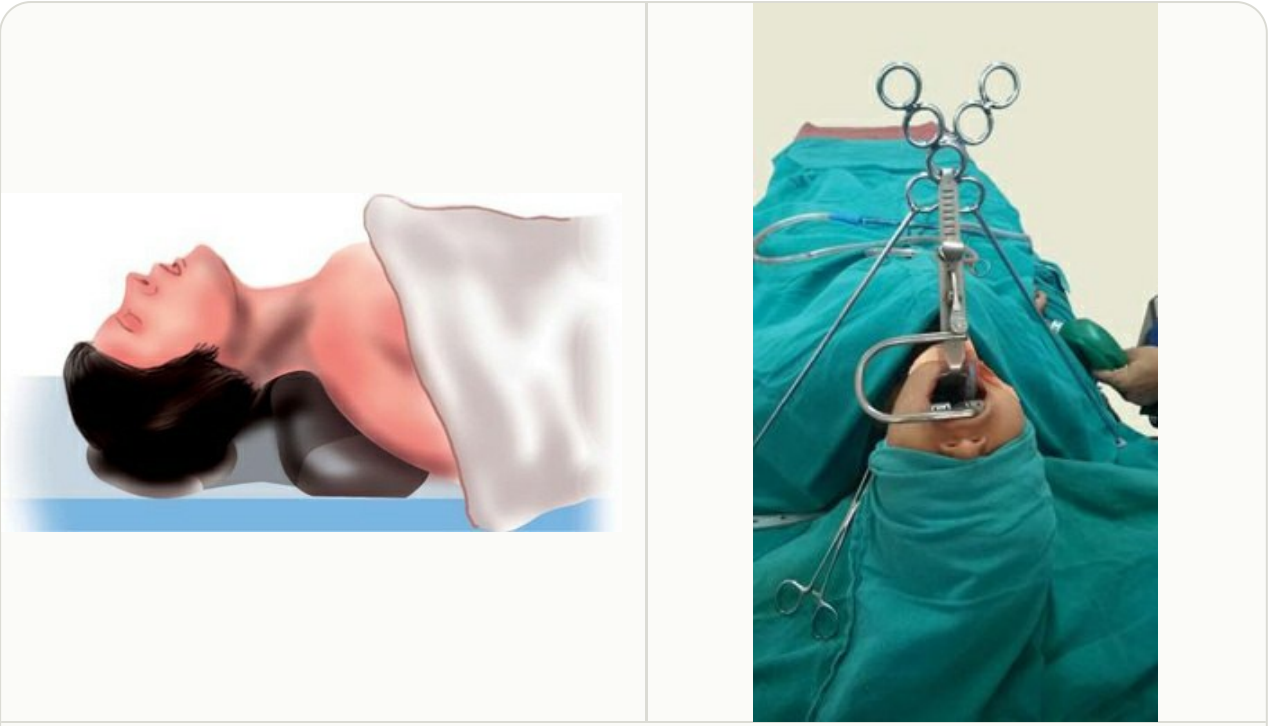


FIGURE 13 First, Rose's position: the neck extended over a sandbag and the head on a ring. Second, the Boyle–Davis gag held by Draffin's bipods.

Dhingra, figs 94.1 and 94.3

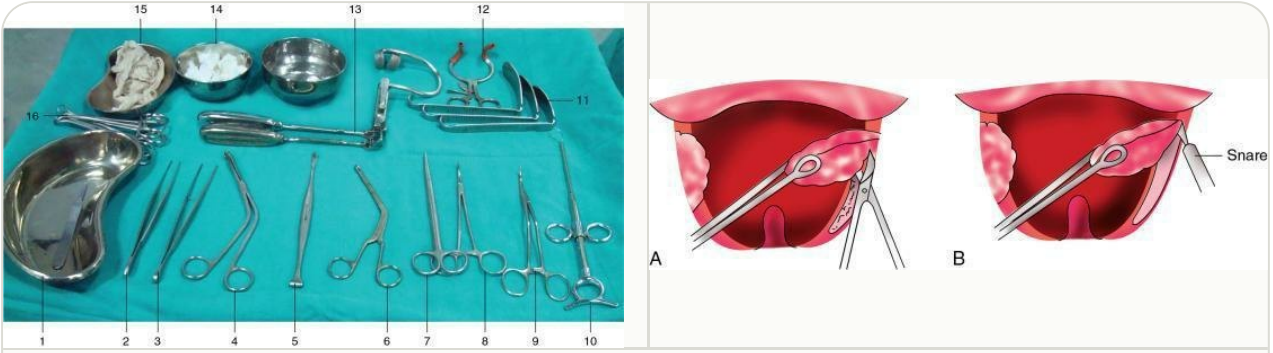


FIGURE 14 First, the instruments for tonsillectomy. Second, (A) the tonsil dissected from its bed and (B) the pedicle at the lower pole cut with a snare.

Dhingra, figs 94.2 and 94.4

6.4 Other techniques

- **Cold:** dissection and snare (the commonest), guillotine (largely abandoned), intracapsular tonsillotomy with a microdebrider (capsule preserved, less pain), harmonic scalpel, coblation (plasma-mediated, low temperature), cryosurgery.
- **Hot:** monopolar or bipolar electrocautery (less bleeding, more pain), KTP or CO2 laser tonsillectomy or staged laser tonsillotomy, radiofrequency.

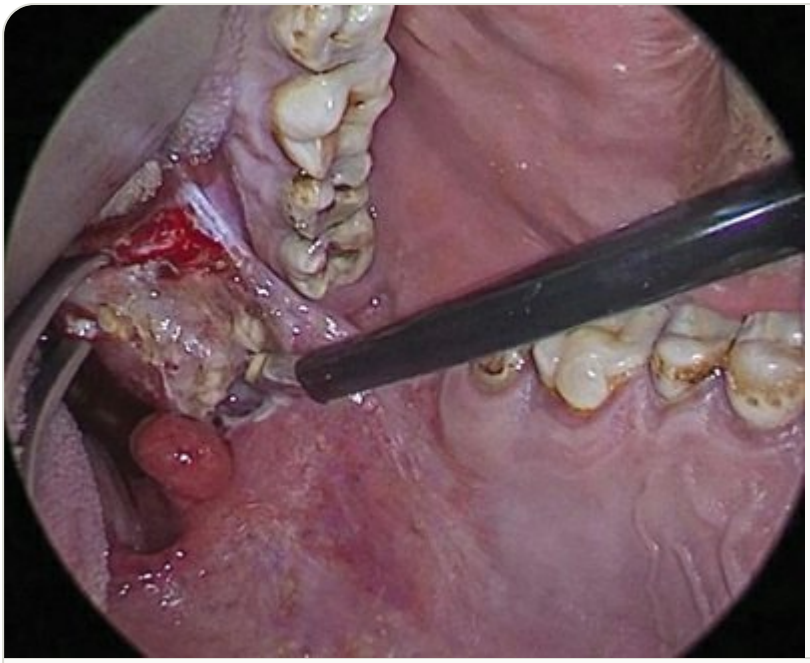


FIGURE 15 Tonsillectomy with coblation.
Dhingra, fig. 94.5

OPERATIVE GUIDES IN THEATRE 2 operations

Tonsillectomy and drainage of neck abscesses.

Tonsillectomy
Tonsillectomy for recurrent criteria or quinsy

Deep Neck Abscesses
Neck abscess

6.5 Aftercare

Recovery position until awake; watch pulse, breathing and blood pressure, and look for bleeding or repeated swallowing. Cold fluids first, then a soft diet, then normal food early, which keeps the fossae clean; regular analgesia before meals (paracetamol, ibuprofen; not aspirin); mouthwashes. Most go home the same or next day and return to work or school in about two weeks. Earache is referred pain, not otitis.

6.6 Complications

6.6.1 Immediate

- **Primary haemorrhage** at operation: pressure, ties, diathermy.

- **Reactionary haemorrhage** within 24 hours: remove the clot (which stops the constrictor from clamping the vessels), apply pressure or vasoconstrictor; if it persists, return to theatre.
- Injury to pillars, uvula, palate, tongue, constrictor or teeth; aspiration of blood; facial oedema; surgical emphysema.

6.6.2 Delayed

- **Secondary haemorrhage** from the fifth to tenth day, from infection and premature separation of the slough, often heralded by bloodstained saliva. Admit every patient; a clot alone is observed, with antibiotics and peroxide gargles; active bleeding returns to theatre under general anaesthesia; check haemoglobin and clotting, transfuse if needed; external carotid ligation is very rarely required.
- Infection of the fossa, parapharyngeal abscess, otitis media.
- Lung complications from aspiration: atelectasis, lung abscess.
- Scarring of the palate and pillars; velopharyngeal insufficiency.
- Tonsillar remnants that become infected; compensatory lingual tonsil hypertrophy.

POST-TONSILLECTOMY BLEEDING

A child who keeps swallowing, vomits dark blood or has a rising pulse is bleeding until proven otherwise. Children compensate until they suddenly do not: secure intravenous access, cross-match, and involve the anaesthetist early; induction in a bleeding child is a high-risk airway.

TECHNIQUE, PAIN AND SAFETY (OXFORD TEXTBOOK)

Techniques are either **extracapsular** (total tonsillectomy, by cold steel, bipolar or monopolar diathermy, or coblation) or **intracapsular** (partial tonsillectomy or tonsillotomy, by coblation or microdebrider, leaving the capsule as a biological dressing over the muscle). Hot techniques reduce operative bleeding but increase secondary haemorrhage and pain, and haemorrhage rises with the diathermy power used, so the lowest effective setting is advised. Intracapsular surgery, especially in young children having surgery for sleep-disordered breathing, gives less pain and fewer bleeds, at the cost of occasional regrowth. After a cluster of child deaths from haemorrhage in Austria, that country moved largely to intracapsular surgery for young children. There is still no consensus on the best method.

Haemorrhage is *primary* within 24 hours (usually from inadequate haemostasis) or *secondary* after 24 hours, most often on days 5 to 10. Every bleed should be assessed in hospital, and any needing intervention counts as significant; reported rates vary widely because definitions differ. Mortality in more than 300,000 English paediatric

tonsillectomies was about 1 in 27,000. Risk is higher in children under three with proven sleep apnoea and in those with craniofacial anomalies, Down syndrome, cerebral palsy, heart disease or bleeding disorders, who need surgery in a centre with paediatric high-dependency care.

Pain is near-universal and causes dehydration and readmission; regular paracetamol with ibuprofen is standard. **Codeine is contraindicated** in under-18s having tonsillectomy for sleep apnoea, because ultrarapid CYP2D6 metabolisers convert it to toxic levels of morphine, causing deaths from respiratory depression. **Adenoidectomy**, done with tonsillectomy in more than half of cases, is now often performed by suction diathermy or coblation under direct or mirror vision rather than with a blind curette.

CHAPTER SEVEN

7 For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Anatomy:** crypta magna, capsule, bed, blood supply and the paratonsillar vein.
- **Types of acute tonsillitis** and complications.
- **Differential diagnosis of a membrane** on the tonsil; **diphtheria** management.
- **Quinsy:** features, site of incision, and when to remove the tonsils.
- **Indications, contraindications** and **complications** of tonsillectomy; reactionary versus secondary haemorrhage.

7.2 Common pitfalls

COMMON PITFALLS

- Draining "quinsy" without trismus or a bulge: aspirate first.
- Giving amoxicillin to a young adult with glandular fever.
- Ignoring a one-sided enlarged tonsil in an adult.

- Sending home a child with a small "warning" bleed after tonsillectomy.

Written for this app from Oxford Textbook of Otolaryngology ch. 49, 87, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 10, Dhingra ch. 51, 52, 94 and K. J. Lee ch. 36, 37, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Tonsillar Abscess

IMAGING FINDINGS

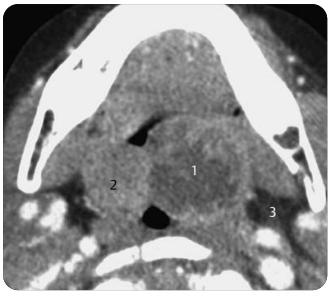
- Rim-enhancing fluid collection in or around the tonsil
- Swollen tonsil and pharyngeal wall
- Spread to parapharyngeal, masticator or carotid space

TOP DIFFERENTIALS

- Tonsillitis without abscess (phlegmon)
- Tonsillar retention cyst
- Retropharyngeal abscess; tonsillar carcinoma

CHECKLIST

- Airway
- Internal jugular vein thrombosis and carotid



CT with intravenous contrast enhancement, axial. The left tonsil (1) is much larger than the contralateral side (2) with central hypodensities…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.